Assignment 1: Cloud Server Health Checker

Scenario

You are working as a **Cloud Infrastructure Engineer**. Your team manages multiple cloud servers and needs a small **CLI tool written in Go** that helps check server health status.

The tool should allow users to **input server details** and check basic information such as CPU usage, memory usage, and security status.

This assignment will help you practice **Go fundamentals such as variables, functions, packages, user input, and modular coding.**

Objective

Build a **Command Line Application in Go** that allows users to:

1. Add server details
2. Display server information
3. Check server health status

Requirements

1. Project Setup

Create a new Go module.

Example:

```
go mod init server-health-checker
```

Project structure should look like:

```
server-health-checker/
│
├── main.go
├── go.mod
│
└── server/
      server.go
```

---

2. Server Structure

Create a **Server struct** containing:

- Server Name
- IP Address
- CPU Usage (%)
- Memory Usage (%)
- Is Secure (true/false)

Example fields:

```
Server Name
IP Address
CPU Usage
Memory Usage
Security Status
```

---

3. Package Creation

Create a **package called server**.

Inside it implement functions like:

1️⃣ CreateServer()

Function that accepts server details and returns a server object.

2️⃣ DisplayServer()

Function that prints server details.

3️⃣ CheckServerHealth()

Logic example:

Condition	Health Status
CPU < 70 AND Memory < 70	Healthy
CPU < 90 AND Memory < 90	Warning
CPU >= 90 OR Memory >= 90	Critical

Return the health status.

---

4. User Input

User should input:

```
Server Name
IP Address
CPU Usage
Memory Usage
Security Status
```

---

5. Output

Display like:

```
------ Server Report ------

Server Name: Auth-Service
```

```
IP Address: 10.10.1.5
CPU Usage: 65%
Memory Usage: 70%
Security Status: Secure

Health Status: WARNING
```

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6. Functions Requirement

Your program must include:

- Functions with parameters
- Functions with return values
- Functions with multiple return values
- Anonymous function (for logging)

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7. Scope Demonstration

Demonstrate **variable scoping**:

- Global variable
- Local variable

Evaluation Criteria

Criteria	Marks
Code Structure	2
Correct Use of Functions	2
Package Usage	2
Input / Output Handling	2
Code Readability	2

Total: **10 Marks**